

Power Query in Excel

In real-world scenarios, data is rarely available in a clean and ready-to-use format. It often contains inconsistencies such as missing values, unnecessary columns, incorrect data types, or data spread across multiple files.

Power Query in Excel is a powerful data transformation tool that enables users to efficiently clean, reshape, and combine data from various sources. It helps automate repetitive data preparation tasks and ensures consistency in data processing.

Definition

Power Query is a data connectivity and transformation tool in Excel that allows users to import, clean, modify, and combine data from multiple sources using a step-by-step process.

Why Do We Need Power Query?

Manual data cleaning can be:

- Time-consuming
- Error-prone
- Repetitive

For example, consider a monthly sales report where:

- Data format changes frequently
- Unwanted rows and columns exist
- Data needs to be combined from multiple files

Using traditional Excel methods, these steps must be repeated every time.

Power Query solves this by allowing users to define the steps once and reuse them with a simple refresh.

Advantages of Power Query

- **Automation:** Once steps are defined, they can be reused by refreshing the data
- **Efficiency:** Handles large datasets more effectively than manual methods
- **Multiple Data Sources:** Supports Excel, CSV, databases, and web data
- **Step-by-Step Transformation:** Each action is recorded and can be modified
- **Error Reduction:** Minimizes manual intervention and mistakes

Limitations of Power Query

- Not suitable for real-time calculations
- Limited compared to programming tools like Python or SQL

- Initial learning curve for beginners
- Complex transformations may impact performance

Where to Find Power Query in Excel

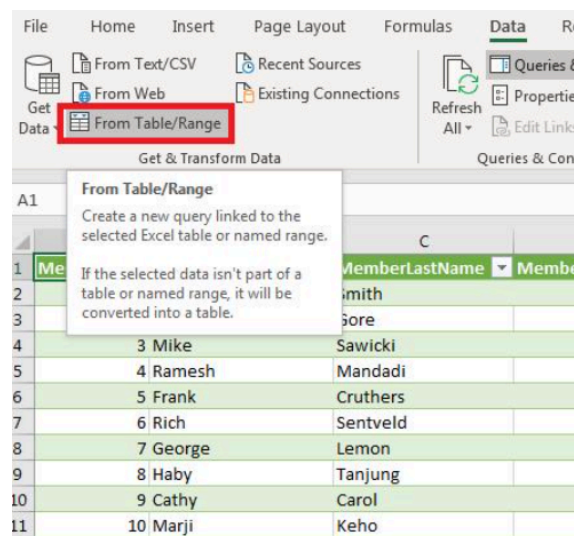
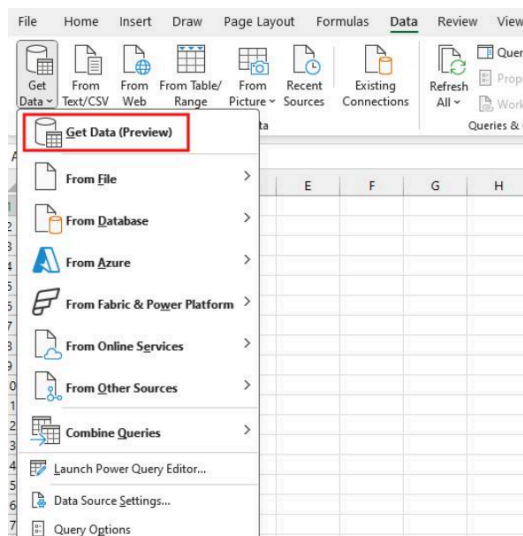
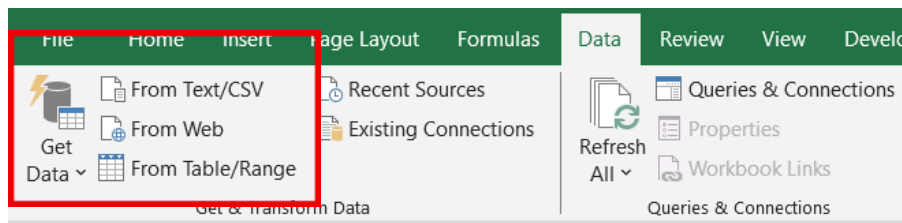
Power Query can be accessed from the Data tab in Excel.

Navigate to:

Data → Get & Transform Data

Common options include:

- Get Data
- From Table/Range
- Queries & Connections



What If Power Query Is Not Available?

Power Query availability depends on the Excel version.

Built-in Availability

- Excel 2016 and later versions: Already included

For Older Versions (Excel 2010 / 2013)

Power Query must be installed as an add-in.

Steps to Enable or Install

1. Go to **File** → **Options** → **Add-ins**
2. Select **COM Add-ins**
3. Click **Go**
4. Enable **Microsoft Power Query for Excel**

If not available, download and install the add-in from Microsoft.

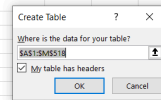
Loading Data into Power Query

Data can be loaded into Power Query in multiple ways:

From Excel Table

1. Convert data into a table (Ctrl + T)

X	Y	month	day	FFMC	DMC	DC	ISI	temp	RH	wind	rain	area
7	5	mar	fri	86.2	26.2	94.3	5.1	8.2	51	6.7	0	0
7	4	oct	tue	90.6	35.4	669.1	6.7	18	33	0.9	0	0
7	4	oct	sat	90.6	43.7	686.9	6.7	14.6	33	1.3	0	0
8	6	mar	fri	91.7	33.3	77.5	9	8.3	97	4	0.2	0
8	6	mar	sun	89.3	51.3	102.2	9.6	11.4	99	1.8	0	0
8	6	aug	sun	92.3	85.3	488	14.7	22.2	29	5.4	0	0
8	6	aug	mon	92.3	88.9	495.6	8.5	24.1	27	3.1	0	0
8	6	aug	mon	91.5	145.4	608.2	10.7	8	86	2.2	0	0
8	6	sep	tue	91	129.5	692.6	7	13.1	63	5.4	0	0
7	5	sep	sat	92.5	88	698.6	7.1	22.8	40	4	0	0
7	5	sep	sat	92.5	88	698.6	7.1	17.8	51	7.2	0	0
7	5	sep	sat	92.8	73.2	713	22.6	19.3	38	4	0	0
6	5	aug	fri	63.5	70.8	665.3	0.8	17	72	6.7	0	0
6	5	sep	mon	90.9	126.5	686.5	7	21.3	42	2.2	0	0
6	5	sep	wed	92.9	133.3	699.6	9.2	26.4	21	4.5	0	0
6	5	sep	fri	93.3	141.2	713.9	13.9	22.9	44	5.4	0	0
5	5	mar	sat	91.7	35.8	80.8	7.8	15.1	27	5.4	0	0
8	5	oct	mon	84.9	32.8	664.2	3	16.7	47	4.9	0	0
6	4	mar	wed	89.2	27.9	70.8	6.3	15.9	35	4	0	0
6	4	apr	sat	86.3	27.4	97.1	5.1	9.3	44	4.5	0	0
6	4	sep	tue	91	129.5	692.6	7	18.3	40	2.7	0	0
5	4	sep	mon	91.8	78.5	724.3	9.2	19.1	38	2.7	0	0
7	4	jun	sun	94.3	96.3	200	56.1	21	44	4.5	0	0
7	4	aug	sat	90.2	110.9	537.4	6.2	19.5	43	5.8	0	0
7	4	aug	sat	93.5	139.4	594.2	20.3	23.7	32	5.8	0	0
7	4	aug	sun	91.4	142.4	601.4	10.6	16.3	60	5.4	0	0



A	B	C	D	E	F	G	H	I	J	K	L	M
X	Y	month	day	FFMC	DMC	DC	ISI	temp	RH	wind	rain	area
7	5	mar	fri	86.2	26.2	94.3	5.1	8.2	51	6.7	0	0
7	4	oct	tue	90.6	35.4	669.1	6.7	18	33	0.9	0	0
7	4	oct	sat	90.6	43.7	686.9	6.7	14.6	33	1.3	0	0
8	6	mar	fri	91.7	33.3	77.5	9	8.3	97	4	0.2	0
8	6	mar	sun	89.3	51.3	102.2	9.6	11.4	99	1.8	0	0
8	6	aug	sun	92.3	85.3	488	14.7	22.2	29	5.4	0	0
8	6	aug	mon	92.3	88.9	495.6	8.5	24.1	27	3.1	0	0
8	6	aug	mon	91.5	145.4	608.2	10.7	8	86	2.2	0	0
8	6	sep	tue	91	129.5	692.6	7	13.1	63	5.4	0	0
7	5	sep	sat	92.5	88	698.6	7.1	22.8	40	4	0	0
7	5	sep	sat	92.5	88	698.6	7.1	17.8	51	7.2	0	0
7	5	sep	sat	92.8	73.2	713	22.6	19.3	38	4	0	0
6	5	aug	fri	63.5	70.8	665.3	0.8	17	72	6.7	0	0
6	5	sep	mon	90.9	126.5	686.5	7	21.3	42	2.2	0	0
6	5	sep	wed	92.9	133.3	699.6	9.2	26.4	21	4.5	0	0
6	5	sep	fri	93.3	141.2	713.9	13.9	22.9	44	5.4	0	0
5	5	mar	sat	91.7	35.8	80.8	7.8	15.1	27	5.4	0	0
8	5	oct	mon	84.9	32.8	664.2	3	16.7	47	4.9	0	0
6	4	mar	wed	89.2	27.9	70.8	6.3	15.9	35	4	0	0
6	4	apr	sat	86.3	27.4	97.1	5.1	9.3	44	4.5	0	0
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7	4	aug	sat	90.2	110.9	537.4	6.2	19.5	43	5.8	0	0
7	4	aug	sat	93.5	139.4	594.2	20.3	23.7	32	5.8	0	0
7	4	aug	sun	91.4	142.4	601.4	10.6	16.3	60	5.4	0	0
7	4	sep	fri	92.4	117.9	668	12.2	19	34	5.8	0	0

2. Go to **Data** → **From Table/Range**

From External Sources

- Excel files
- CSV/Text files
- Databases

- Web sources

Real-Life Example

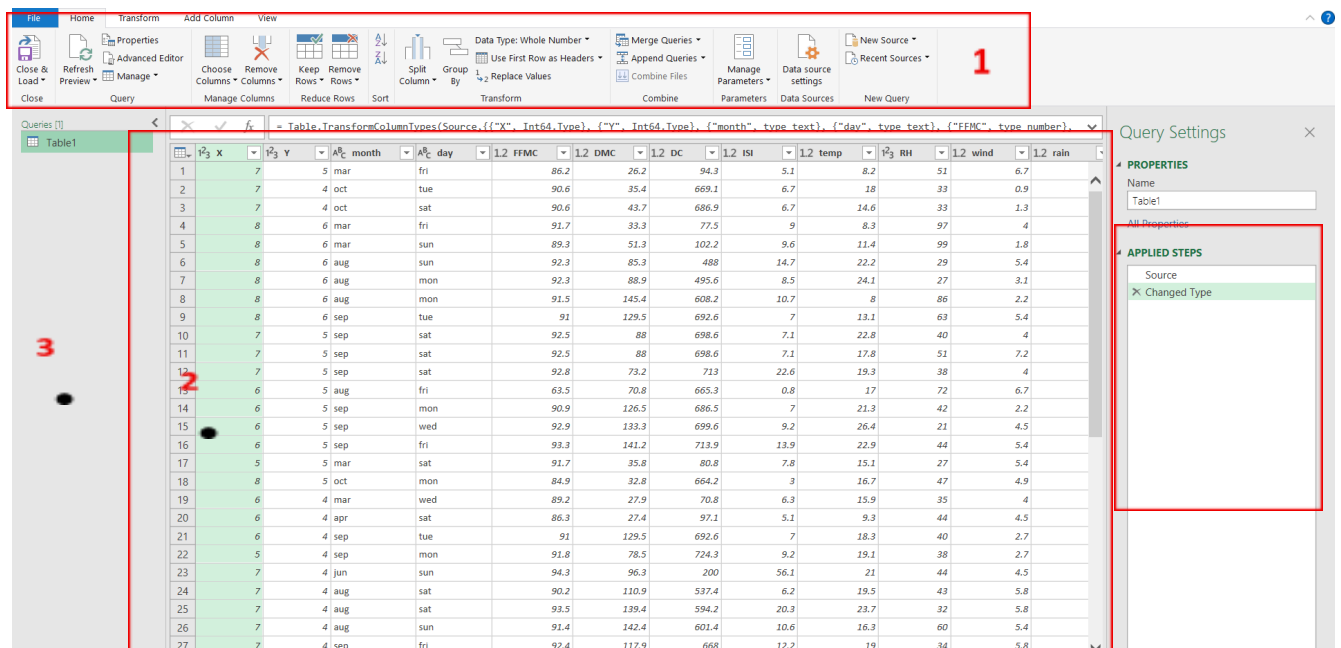
Import sales data from:

- Region 1 Excel
- Region 2 CSV
- Region 3 database

Combine everything into one dataset

Power Query Interface Overview

When you open Power Query Editor, you'll see:



1. Ribbon

Contains transformation tools:

- Remove rows
- Split column
- Merge queries

2. Data Preview

Shows your dataset

3. Applied Steps (Very Important)

This is the heart of Power Query.

- Every action is recorded step-by-step
- Can be edited, deleted, or reordered

Core Operations in Power Query

1. Data Cleaning

- Removing unnecessary columns
- Removing blank or unwanted rows
- Changing data types

2. Data Transformation

- Splitting columns (e.g., full name into first and last name)
- Merging columns
- Replacing values

3. Data Enrichment

- Adding conditional columns
- Creating custom columns

4. Data Aggregation

- Grouping data (Sum, Count, Average)

5. Data Integration

- Merging queries (similar to joins in SQL)
- Appending queries (combining datasets vertically)

Close and Load

After completing all transformations:

Click **Close & Load** to export the processed data back to Excel.

Refreshing Data (Automation Feature)

One of the key benefits of Power Query is automation.

Once transformations are defined:

- Replace or update the source data
- Click **Refresh**

All steps are automatically reapplied

Note: **Power Query transformations do not modify the original source data; all changes are applied only to the loaded output.**

Performing Data Cleaning in Power Query

The following steps demonstrate common data cleaning and transformation operations in Power Query.

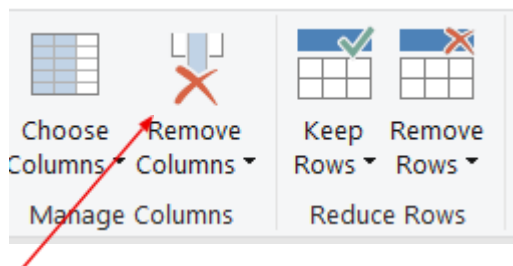
1. Removing Columns

Purpose:

To eliminate unnecessary columns and retain only relevant data.

Steps:

1. Select the column(s) you want to remove
2. Go to the **Home** tab
3. Click **Remove Columns**



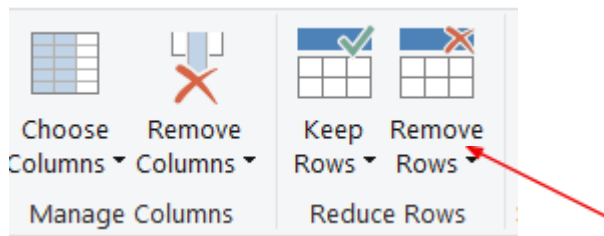
2. Removing Rows

Purpose:

To remove unwanted or blank rows from the dataset.

Steps:

1. Go to the **Home** tab
2. Click **Remove Rows**
3. Choose the required option (Top, Bottom, or Blank rows)



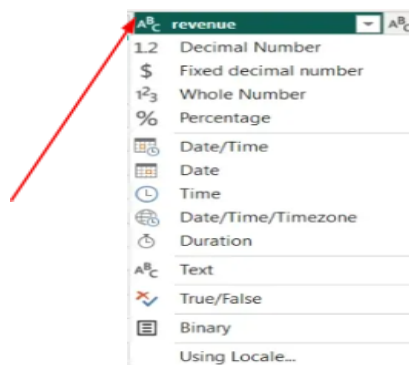
3. Changing Data Type

Purpose:

To assign the correct data format to each column.

Steps:

1. Select the column
2. Click the data type icon in the column header
3. Choose the appropriate data type



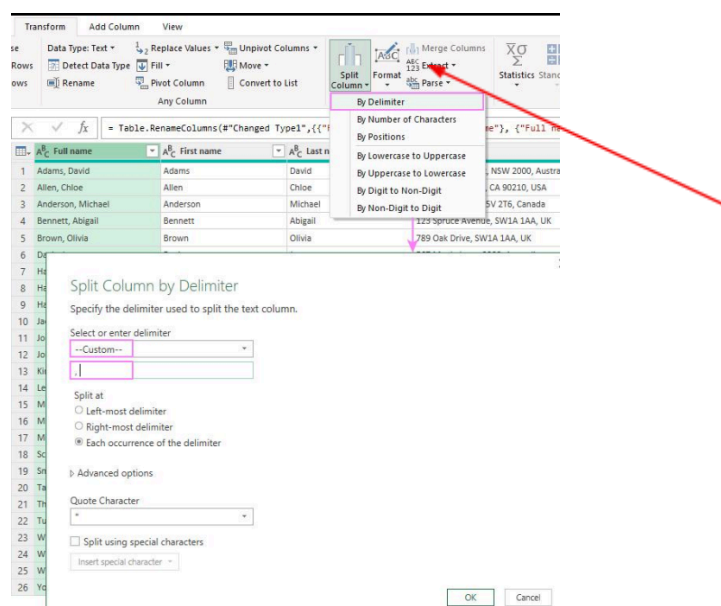
4. Splitting Columns

Purpose:

To divide a column into multiple columns based on a delimiter.

Steps:

1. Select the column
2. Go to the **Transform** tab
3. Click **Split Column** → **By Delimiter**
4. Select the delimiter and apply



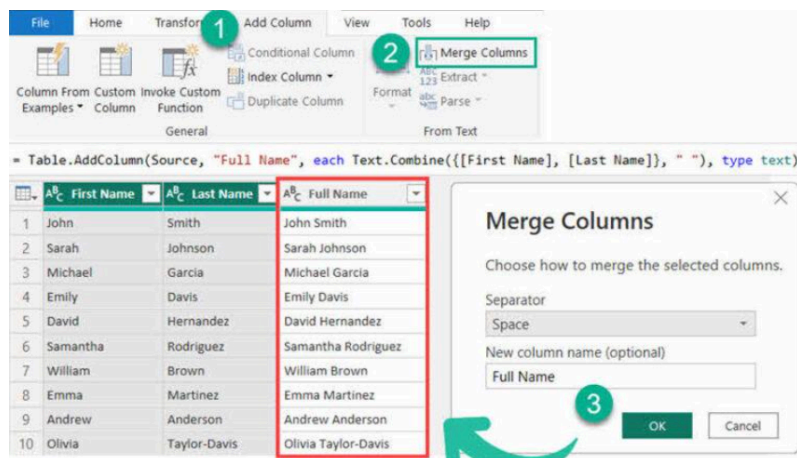
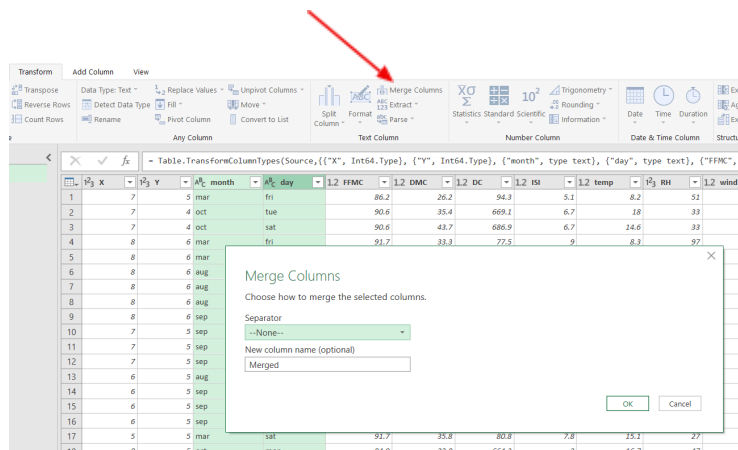
5. Merge Columns

Purpose:

To combine multiple columns into a single column.

Steps:

1. Select multiple columns (Ctrl + Click)
2. Go to **Transform** tab
3. Click **Merge Columns**
4. Select separator and confirm



6. Replace Values

Purpose:

To modify or standardize specific values within a column.

Steps:

1. Select the column
2. Go To Transform Tab
3. Click Replace values

4. Enter value to find and replace

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

OK Cancel

7. Add Conditional Column

Purpose:

To create a new column based on defined conditions or logic.

Steps:

1. Go to **Add Column** tab
2. Click **Conditional Column**
3. Define condition and output

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Custom

Column Name Operator Value Output

If FFMC > 125 Then 125

Add Clause

Ease

125

OK Cancel

Add Conditional Column
Add a conditional column that is computed from the other columns or values.

New column name
Donor Level

	Column Name	Operator	Value	Output
If	Donation	is greater than or...	4000	Platinum
Else If	Donation	is greater than or...	3000	Gold
Else If	Donation	is greater than or...	2000	Silver

Add Clause

Else
Bronze

OK Cancel

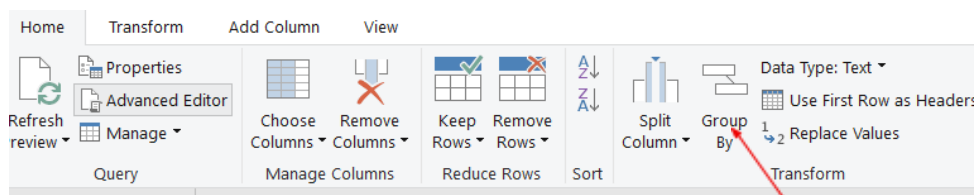
8. Group By

Purpose:

To summarize data by aggregating values based on a selected column.

Steps:

1. Select the column
2. Go to **Home** → **Group By**
3. Choose aggregation (Sum, Count, Avg)
4. Click OK



Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

wind

New column name

Count

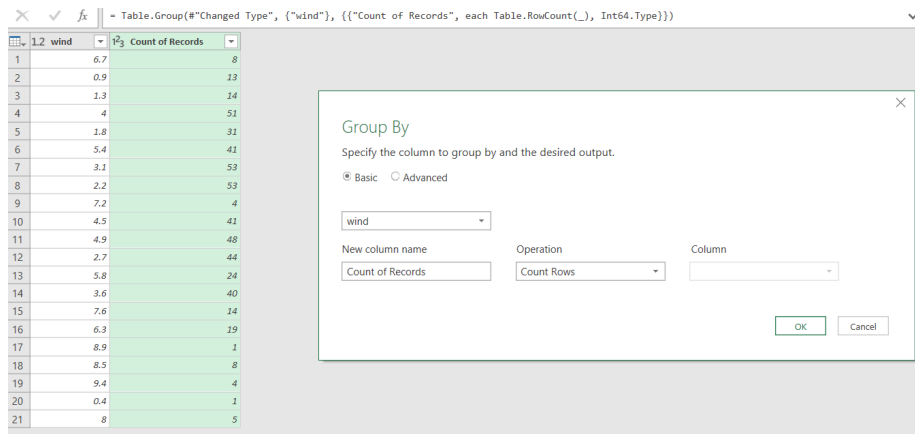
Operation

Count Rows

Column

OK

Cancel



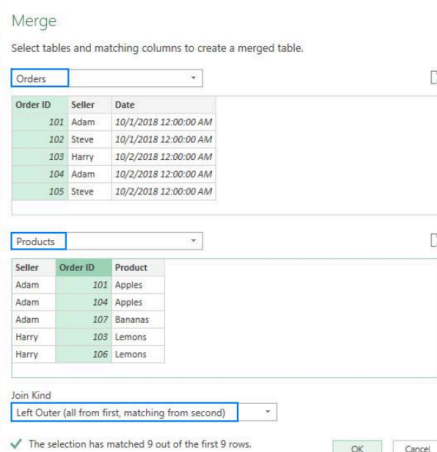
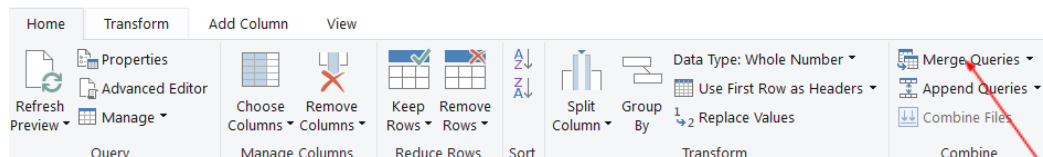
9. Merge Queries

Purpose:

To combine data from multiple tables using a common column.

Steps:

1. Go to **Home** → **Merge Queries**
2. Select both tables
3. Choose common column
4. Select join type
5. Click OK



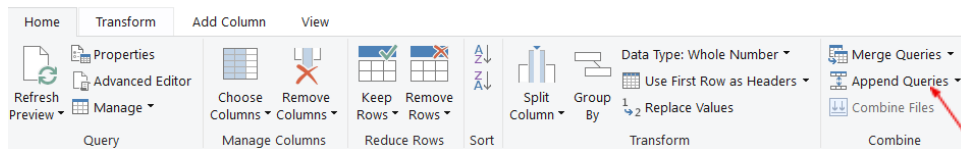
10. Append Queries

Purpose:

To combine multiple tables by stacking rows vertically.

Steps:

1. Go to **Home** → **Append Queries**
2. Select tables to combine
3. Click OK



Append ?

Concatenate rows from two tables into a single table.

☒ Two tables ☐ Three or more tables

Table to append *

Store Sales

OK Cancel

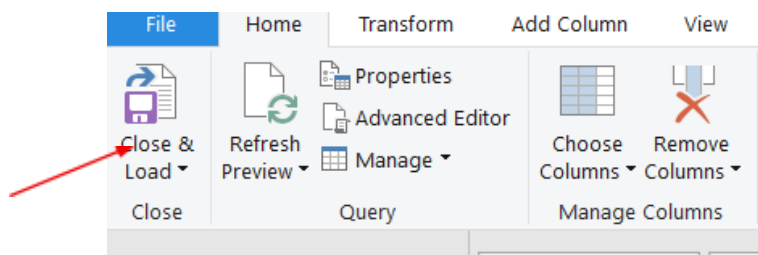
11. Close & Load

Purpose:

To load the transformed data from Power Query back into Excel.

Steps:

1. Click **Close & Load**
2. Select output option



12. Refresh Data

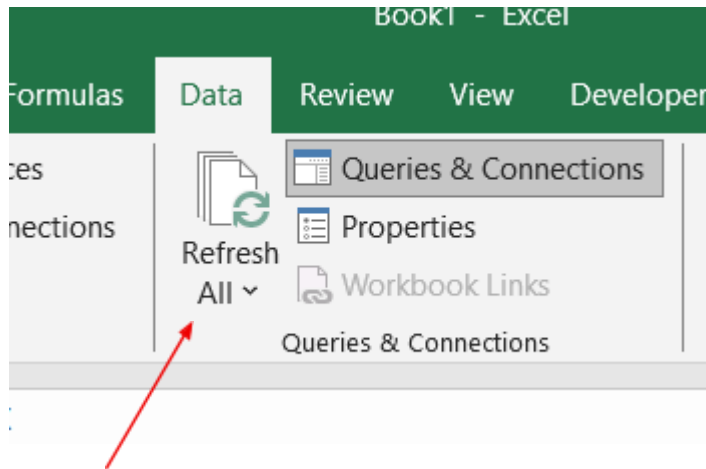
Purpose:

To update the output by reapplying all transformation steps on the latest data.

Steps:

1. Update source data

2. Go to **Data** tab
3. Click **Refresh**



Power Query enables users to efficiently clean, transform, and automate data preparation processes, making it an essential tool for handling real-world datasets.